

Robust Adversarial Quantification VIA Conflict-Aware Evidential Deep Learning

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Threats; 2 Ways a Model Gets Blindsided:



Out of Distribution (OOD):

Out Of Distribution data is when a model sees something it wasn't trained on. E. g., a digit classifier shown a handwritten letter.

Nothing malicious, the world is bigger than the training set.



Adversarial Inputs:

Adversarial Inputs are inputs that were deliberately modified to perturb and confuse the model. They are often invisible to the human eye since they target the mathematical properties of models.

Problem; Silent Failure Is the Real Danger:



Standard classifiers are the answer

Background check from the first EDL paper.

- Instead of forcing a single probability vector out of the network, EDL asks the network to output evidence ($e_k \geq 0$). If the network has no basis for any class, it can output $e = [0, 0, 0]$ which can be translated as a structurally different, honestly-reported "I don't know," rather than a softmax vector that's forced to look like an opinion no matter what.

That silent commitment is exactly what adversarial attacks exploit...



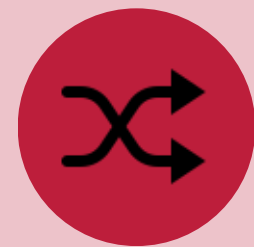
Dangers of silent overconfidence:

- Autonomous driving
Misreading a stop sign as a speed limit sign with full confidence
- Medical diagnosis
A scan the model has never seen classified as routine
- Fraud / Security systems
A crafted input sails through as clean

Any system that abstains more will also reject some good inputs. Every result in this paper lives inside that trade-off.

Foundation:

Two kinds of “I-Don’t-Knows”



Aleatoric Uncertainty:

Irreducible noise in the data itself. A blurry photo will always be ambiguous. The confusion lives in the sample itself.

Not this paper’s focus



Epistemic Uncertainty:

Uncertainty from the model's limited knowledge. In contrast to the aleatoric, more training data helps reduce this.

This paper’s main focus

The Math: EDL Formalized:

Let $X \subseteq \mathbb{R}^d$ denote the input space and $Y = \{1, \dots, K\}$ the label space with K classes.:

$\alpha_k = \mathbf{e}_k + 1$: Dirichlet parameter from per-class evidence $e_k \geq 0$.

$\mathbf{S} = \sum \alpha_k$: Total Dirichlet strength across all K classes.

$\mathbf{b}_k = (\alpha_k - 1) / \mathbf{S}$: Belief mass assigned to class k .

$\mathbf{u} = K / \mathbf{S}$: Uncertainty mass, shrinks as evidence grows.

The Math: EDL Formalized:

The Dirichlet distribution itself is expressed as the following:

$$\text{Dir}(p \mid \alpha) = \frac{\Gamma(S)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K p_k^{\alpha_k - 1}$$

$p = (p_1, p_2, \dots, p_k)$ is a vector of probabilities for K different categories.

$\Gamma(x)$ is the gamma function, basically $\Gamma(n) = (n-1)!$

$\frac{\Gamma(S)}{\prod_{k=1}^K \Gamma(\alpha_k)}$ Normalization constant

$\prod_{k=1}^K p_k^{\alpha_k - 1}$ Shape core, determines the shape of the distribution itself

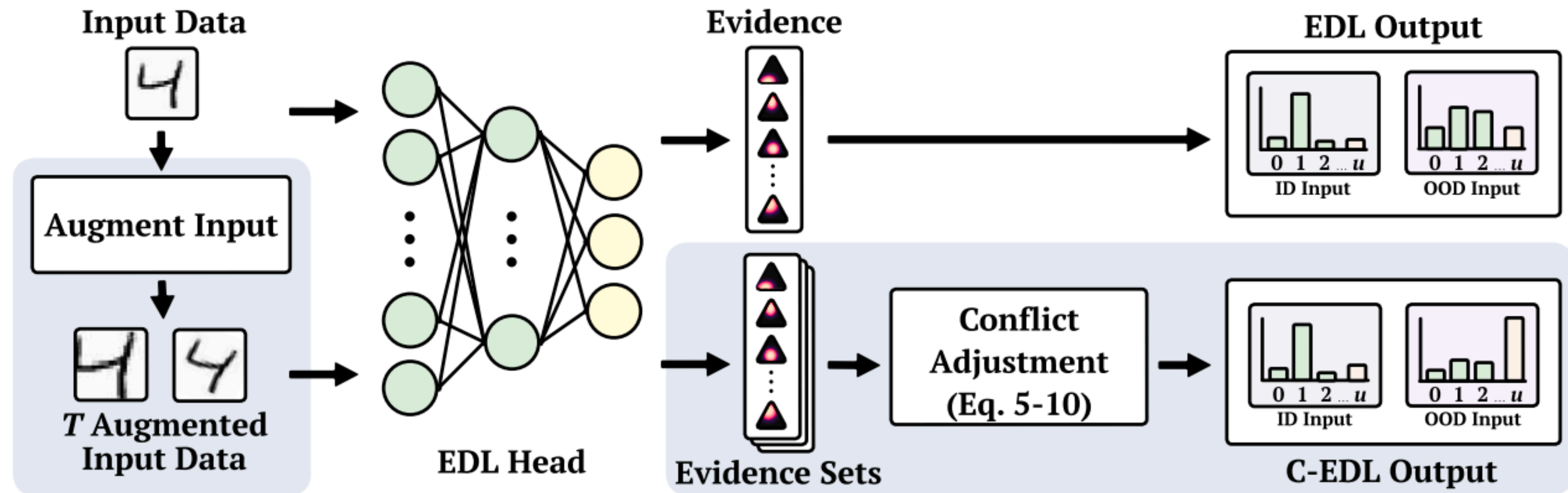
The Math: EDL Formalized:

The Dirichlet distribution itself is expressed as the following:

$$\text{Dir}(p \mid \alpha) = \frac{\Gamma(S)}{\prod_{k=1}^K \Gamma(\alpha_k)} \prod_{k=1}^K p_k^{\alpha_k - 1}$$

- A high Dirichlet (large α_k , large S) means the model is confident about which point in the class probability vector is right.
- A flat Dirichlet ($S \approx K$) means high uncertainty, the model is unsure which p is plausible

C-EDL Overview:



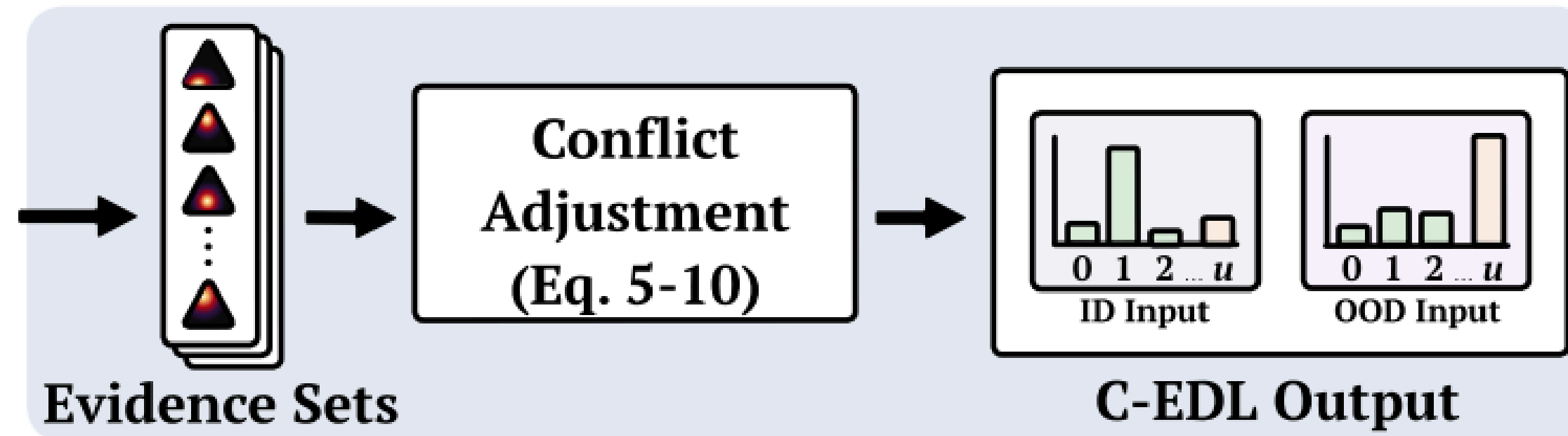
- C-EDL is a post-hoc method on top of the traditional EDL.
- Its key methods are highlighted in blue.

Input Augmentation:

In order to diversify the set of evidence for a given input, C-EDL applies a set of T metamorphic transformations $\{\tau_1, \tau_2, \dots, \tau_T\}$

- Each $\{\tau_t(x)\}$ serves as a distinct but semantically equivalent view of the original input.
- A model that has learned robust decision features should produce consistent evidence across all transformations.
- This property enables C-EDL to probe the stability of the model's beliefs under controlled, task-instrumented perturbations.
- Every transformation generates a corresponding Dirichlet vector $\alpha^{(t)}$ that encodes the strength belief.

Conflict Adjustment:



After collecting the diverse evidence sets, we quantify disagreement across these views through two complementary measures of conflict.

- Intra-class variability
- Inter-class contradiction

Conflict Adjustment; Intra-class Variability:

$$C_{intra} = \frac{1}{K} \sum_{k=1}^K \frac{\sigma(\{\alpha_k^{(t)}\}_{t=1}^T)}{\mu(\{\alpha_k^{(t)}\}_{t=1}^T) + \epsilon}$$

Intra-class variability captures how much the evidence for each class fluctuates across the applied transformations.

- $\sigma(\cdot)$ is the standard deviation, “The Spread”.
- $\mu(\cdot)$ is the mean, “The Average”.
- ϵ is a small positive constant used for numerical stability.

Conflict Adjustment; Inter-class Contradiction:

$$C_{inter} = \frac{1}{T} \sum_{t=1}^T \left(1 - \exp \left(-\beta \sum_{k=1}^K \sum_{j=k+1}^K \left(\frac{\min(\alpha_k^{(t)}, \alpha_j^{(t)})}{\max(\alpha_k^{(t)}, \alpha_j^{(t)})} \times \frac{\min(\alpha_k^{(t)}, \alpha_j^{(t)})}{\sum_{k=1}^K \alpha_k^{(t)}} \times 2 \right)^2 \right) \right)$$

Inter-class conflict measures instances where the model supports competing classes, highlighting cases where the model is unsure about its prediction.

- $\beta > 0$ is a scaling parameter that adjusts the sharpness of the penalty assigned to the contradiction
- the ratio term captures the relative balance between classes while the strength normalization penalises contradictions arising from uniformly low evidence.

Conflict Adjustment; “Final” Conflict-Score:

To obtain the final measurement of conflict, both terms are combined into a single total score:

$$C = C_{\text{inter}} + C_{\text{intra}} - C_{\text{inter}}C_{\text{intra}} - \lambda(C_{\text{inter}} - C_{\text{intra}})^2$$

Using the inclusion-exclusion principle, which ensures a combined measure of conflict by accounting for overlap between both inter-class and intra-class measures:

- $\lambda \in [0, 1]$ controls the penalisation of asymmetric disagreement.
- This formulation ensures $C \in (0, 1]$ tends towards 0 if and only if all transformations produce identical Dirichlet parameters

Conflict Adjustment; Post-hoc Uncertainty Augmentation:

The obtained Conflict-score serves as the basis of the post-hoc uncertainty augmentation and evidence reduction of C-EDL:

$$\bar{\alpha}_k = \frac{1}{T} \sum_{t=1}^T \alpha_k^{(t)}$$

Aggregation of all the Dirichlet parameters

$$\tilde{\alpha}_k = \bar{\alpha}_k \times \exp(-\delta C)$$

Exponential decay scaling every parameter to reduce overconfidence

$\delta > 0$ is a tunable hyperparameter controlling the sensitivity of the adjustment

Conflict Adjustment; Post-hoc Uncertainty Augmentation:

Finally, the remaining C-EDL calculations:

$$\tilde{S} = \sum_{k=1}^K \tilde{\alpha}_k, \quad \tilde{b}_k = \frac{\tilde{\alpha}_k - 1}{\tilde{S}}, \quad \tilde{u} = \frac{K}{\tilde{S}}, \quad \mathbb{E}[\tilde{p}_k] = \frac{\tilde{\alpha}_k}{\tilde{S}}.$$

Total Dirichlet strength,
reduced when C is
high.

Belief per class.

Uncertainty mass.

Expected
probabilities.

Evaluation:

In order to evaluate C-EDL's ability, it is compared to other state-of-the-art EDL-based models and other UQ approaches through a series of comprehensive tests and over 10 independent runs.

In the next few slides, said networks and the used datasets will be explained along the comparison of the results.

Evaluation; Competitive Approach:

Model	Inner workings	Metaphorical example
Posterior Network	Uses DD to map input to a lower dimensional space and learn features	Ignore the witness's confidence; check the case-file archive directly.
Evidential Deep Learning	Network outputs non-negative "evidence" per class	States their evidence once; you take it at face value.
Fisher Information-EDL	Reweights loss using the Fisher information so evidence that's less "informative" is penalized less	Weighted by how reliable this witness tends to be on this topic.
Smoothed-EDL	Regularizes predictions to stay stable under small input perturbations	Asked the same question several ways; you take the median answer.
Hyper Opinion-EDL	Assigns belief to subsets of classes, not just single classes	Lets the witness name a group of suspects, not just one.
Density Aware-EDL	Scales evidence by local feature-space density	Cross-checked against how similar this case is to solved ones.
Relaxed-EDL	Makes the prior-weight parameter tweakable and removes the variance-minimizing term	The bar for "confident" testimony is loosened.

Evaluation; Component Effect:

In order to study each C-EDL components, three variants alongside C-EDL were evaluated to isolate their specific impacts:

- Metamorphic Transformation was replaced with Monte Carlo dropout.
- Conflict-Awareness was ruled out for naive averaging.

	Naive averaging	Conflict-Aware
MC Dropout	EDL++ (MC)	C-EDL (MC)
Metamorphic Transformations	EDL++	C-EDL (Meta)

Evaluation; Datasets:



MNIST



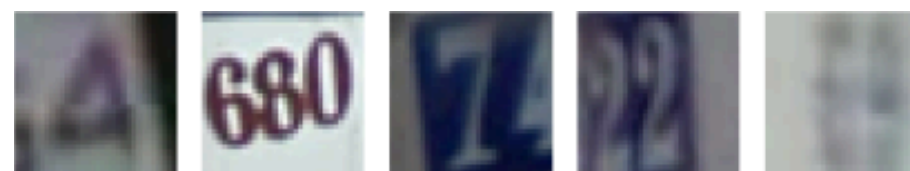
CIFAR-10



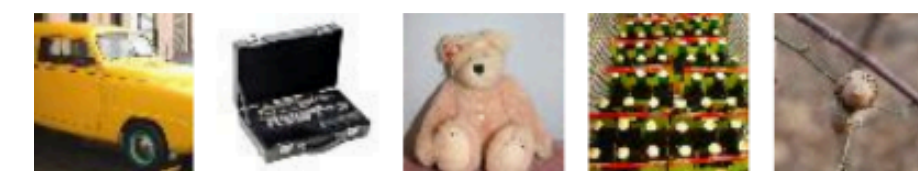
Deep Weeds



FMNIST



SVHN



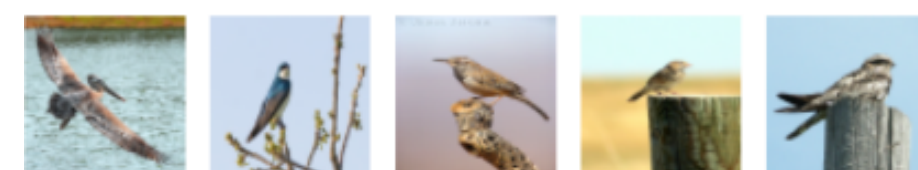
Tiny-ImageNet



KMNIST



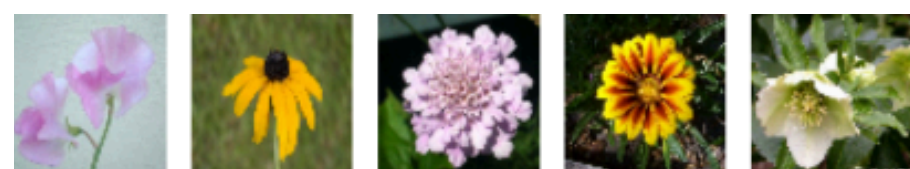
CIFAR-100



Caltech-UCSD Birds-200-2011 (CUB)



EMNIST



Oxford Flowers

Evaluation; Far-OOD vs. Near-OOD:

Far-OOD:

Very far from training data and are very obvious to a human. Models usually give a low confidence score, e.g.:



VS



Near-OOD:

semantically related to the training data and share many visual features. Difficult to detect and models often predict these with a high confidence, e. g.:



VS



Evaluation; Core Results:

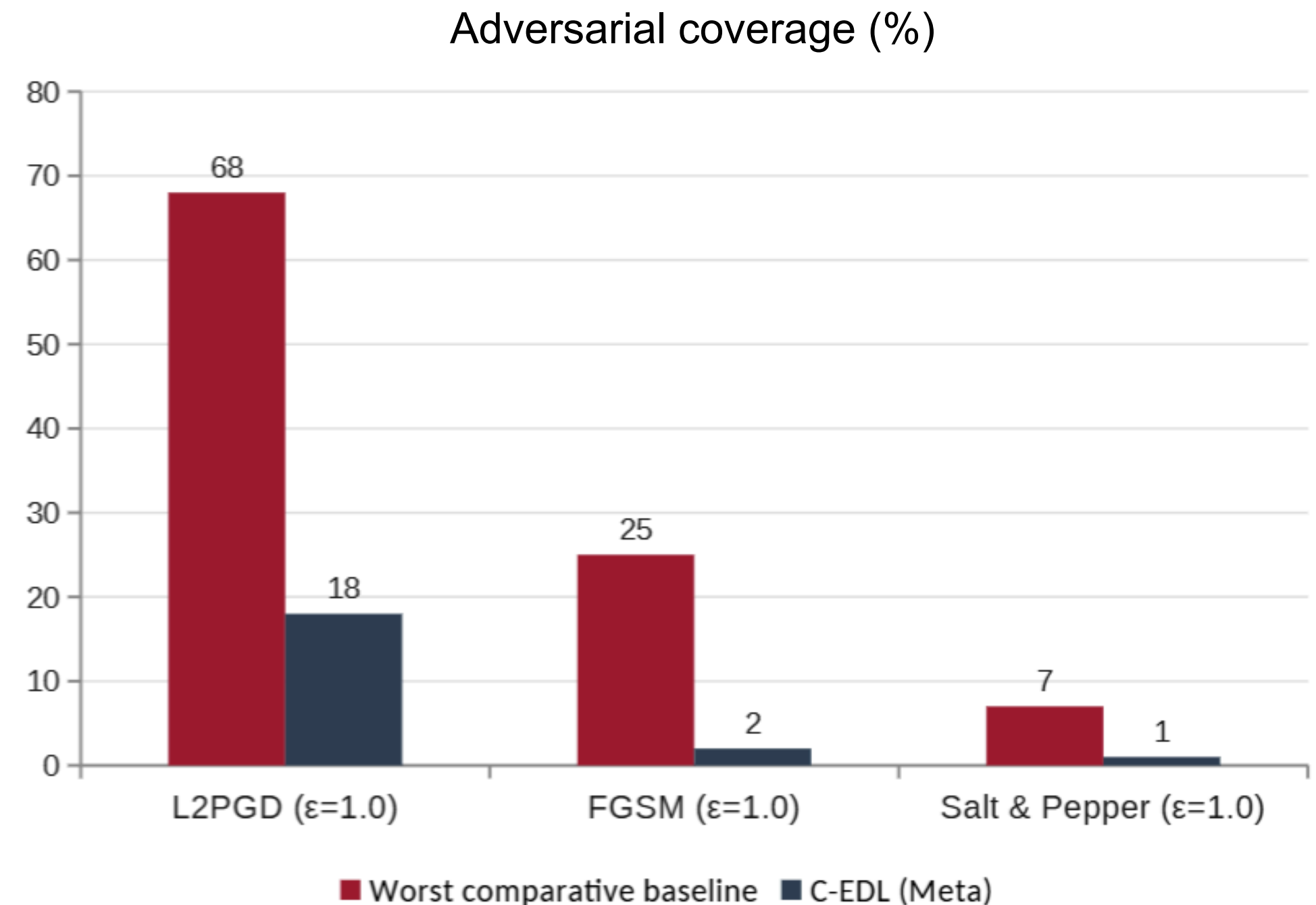
	Comparative Methods							Ablated C-EDL Methods			Proposed C-EDL
	Posterior Network	EDL	$\mathcal{I}$ -EDL	S-EDL	H-EDL	R-EDL	DA-EDL	EDL++ (MC)	C-EDL (MC)	EDL++ (Meta)	C-EDL (Meta)
MNIST $\rightarrow$ FashionMNIST											
ID Acc (%) $\uparrow$	99.96 $\pm$ 0.01	99.96 $\pm$ 0.02	99.95 $\pm$ 0.02	99.95 $\pm$ 0.02	99.96 $\pm$ 0.02	99.96 $\pm$ 0.01	99.89 $\pm$ 0.05	99.97 $\pm$ 0.01	99.98 $\pm$ 0.02	99.97 $\pm$ 0.02	99.96 $\pm$ 0.01
ID Cov (%) $\uparrow$	94.72 $\pm$ 0.96	96.61 $\pm$ 0.57	96.08 $\pm$ 0.67	96.06 $\pm$ 0.72	96.18 $\pm$ 0.92	96.27 $\pm$ 0.48	95.16 $\pm$ 0.76	93.35 $\pm$ 1.28	92.14 $\pm$ 1.43	94.93 $\pm$ 0.60	94.18 $\pm$ 1.03
OOD Cov (%) $\downarrow$	3.55 $\pm$ 0.67	2.52 $\pm$ 0.68	2.74 $\pm$ 0.96	2.41 $\pm$ 0.66	2.28 $\pm$ 0.98	2.34 $\pm$ 0.86	2.98 $\pm$ 2.02	7.55 $\pm$ 1.33	1.77 $\pm$ 0.63	1.96 $\pm$ 0.59	2.00 $\pm$ 0.80
Adv Cov (%) $\downarrow$	61.14 $\pm$ 9.38	52.21 $\pm$ 9.49	47.58 $\pm$ 8.79	48.80 $\pm$ 8.75	50.40 $\pm$ 14.60	49.05 $\pm$ 11.90	28.74 $\pm$ 7.93	38.81 $\pm$ 4.23	21.62 $\pm$ 4.89	16.43 $\pm$ 5.52	15.51 $\pm$ 6.09
MNIST $\rightarrow$ KMNIST											
ID Acc (%) $\uparrow$	99.97 $\pm$ 0.01	99.97 $\pm$ 0.01	99.96 $\pm$ 0.01	99.97 $\pm$ 0.01	99.97 $\pm$ 0.01	99.96 $\pm$ 0.01	99.90 $\pm$ 0.03	99.98 $\pm$ 0.01	99.98 $\pm$ 0.02	99.98 $\pm$ 0.01	99.98 $\pm$ 0.01
ID Cov (%) $\uparrow$	94.72 $\pm$ 0.51	95.42 $\pm$ 0.62	95.47 $\pm$ 0.55	95.08 $\pm$ 0.78	95.59 $\pm$ 0.48	95.32 $\pm$ 0.53	94.12 $\pm$ 1.20	91.94 $\pm$ 0.99	90.20 $\pm$ 1.66	93.73 $\pm$ 0.82	92.42 $\pm$ 0.65
OOD Cov (%) $\downarrow$	3.78 $\pm$ 0.81	3.23 $\pm$ 0.76	3.54 $\pm$ 0.74	3.16 $\pm$ 0.56	3.33 $\pm$ 0.59	3.21 $\pm$ 0.37	3.37 $\pm$ 2.35	11.67 $\pm$ 1.19	3.69 $\pm$ 1.01	2.21 $\pm$ 0.58	1.90 $\pm$ 0.32
Adv Cov (%) $\downarrow$	23.91 $\pm$ 3.70	20.88 $\pm$ 5.95	19.57 $\pm$ 4.27	14.96 $\pm$ 4.11	20.03 $\pm$ 2.91	16.40 $\pm$ 4.49	10.06 $\pm$ 4.21	27.79 $\pm$ 2.66	12.40 $\pm$ 2.09	4.20 $\pm$ 1.93	3.01 $\pm$ 0.94
MNIST $\rightarrow$ EMNIST*											
ID Acc (%) $\uparrow$	99.98 $\pm$ 0.01	99.98 $\pm$ 0.01	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	99.98 $\pm$ 0.01	99.98 $\pm$ 0.02	99.92 $\pm$ 0.04	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	99.99 $\pm$ 0.01	99.98 $\pm$ 0.01
ID Cov (%) $\uparrow$	93.14 $\pm$ 0.87	92.73 $\pm$ 0.94	93.17 $\pm$ 0.46	92.65 $\pm$ 1.10	93.18 $\pm$ 1.16	93.27 $\pm$ 0.82	90.53 $\pm$ 1.30	86.11 $\pm$ 2.86	82.75 $\pm$ 2.96	90.65 $\pm$ 1.34	89.89 $\pm$ 1.44
OOD Cov (%) $\downarrow$	14.96 $\pm$ 1.02	10.74 $\pm$ 1.34	10.62 $\pm$ 0.93	10.31 $\pm$ 1.33	11.29 $\pm$ 1.59	10.06 $\pm$ 1.02	14.14 $\pm$ 8.47	17.16 $\pm$ 1.34	6.83 $\pm$ 1.37	8.88 $\pm$ 1.04	8.93 $\pm$ 0.75
Adv Cov (%) $\downarrow$	15.76 $\pm$ 3.95	7.81 $\pm$ 3.53	6.53 $\pm$ 1.72	6.60 $\pm$ 2.02	9.05 $\pm$ 4.11	6.40 $\pm$ 2.96	12.30 $\pm$ 10.93	22.63 $\pm$ 2.74	7.14 $\pm$ 1.61	1.47 $\pm$ 0.59	1.41 $\pm$ 0.40
CIFAR10 $\rightarrow$ SVHN											
ID Acc (%) $\uparrow$	95.63 $\pm$ 0.72	95.88 $\pm$ 0.54	96.19 $\pm$ 0.70	96.07 $\pm$ 0.69	96.09 $\pm$ 0.55	96.82 $\pm$ 0.54	91.25 $\pm$ 0.96	97.39 $\pm$ 0.53	98.40 $\pm$ 0.39	97.29 $\pm$ 0.31	97.98 $\pm$ 0.42
ID Cov (%) $\uparrow$	65.74 $\pm$ 3.11	67.34 $\pm$ 3.15	66.57 $\pm$ 2.41	65.57 $\pm$ 3.36	66.74 $\pm$ 2.75	62.59 $\pm$ 2.41	43.83 $\pm$ 16.97	56.65 $\pm$ 3.01	47.04 $\pm$ 3.64	60.53 $\pm$ 2.32	54.70 $\pm$ 2.11
OOD Cov (%) $\downarrow$	11.82 $\pm$ 2.52	10.91 $\pm$ 2.23	9.68 $\pm$ 1.84	10.40 $\pm$ 1.94	9.84 $\pm$ 1.63	8.92 $\pm$ 1.76	20.40 $\pm$ 9.25	12.01 $\pm$ 1.88	6.66 $\pm$ 1.06	6.59 $\pm$ 1.17	4.69 $\pm$ 1.00
Adv Cov (%) $\downarrow$	15.57 $\pm$ 6.19	20.00 $\pm$ 6.89	19.25 $\pm$ 6.37	3.32 $\pm$ 1.46	19.64 $\pm$ 4.86	14.61 $\pm$ 6.79	21.48 $\pm$ 8.98	15.22 $\pm$ 2.32	9.39 $\pm$ 1.18	2.35 $\pm$ 0.96	1.25 $\pm$ 0.56
CIFAR10 $\rightarrow$ CIFAR100*											
ID Acc (%) $\uparrow$	96.17 $\pm$ 0.36	96.66 $\pm$ 0.70	96.67 $\pm$ 0.45	96.69 $\pm$ 0.54	96.76 $\pm$ 0.46	97.40 $\pm$ 0.41	89.28 $\pm$ 0.81	97.87 $\pm$ 0.28	98.64 $\pm$ 0.40	97.51 $\pm$ 0.40	98.22 $\pm$ 0.27
ID Cov (%) $\uparrow$	62.91 $\pm$ 2.71	63.33 $\pm$ 3.89	63.84 $\pm$ 2.75	63.13 $\pm$ 2.35	62.25 $\pm$ 2.11	60.21 $\pm$ 2.39	66.98 $\pm$ 3.51	54.50 $\pm$ 2.37	45.02 $\pm$ 3.37	59.47 $\pm$ 2.31	53.18 $\pm$ 2.31
OOD Cov (%) $\downarrow$	17.73 $\pm$ 2.42	17.24 $\pm$ 4.01	17.73 $\pm$ 2.72	17.48 $\pm$ 2.56	16.12 $\pm$ 2.01	14.46 $\pm$ 2.02	33.85 $\pm$ 4.14	19.50 $\pm$ 1.74	11.44 $\pm$ 1.81	13.34 $\pm$ 1.83	9.61 $\pm$ 1.26
Adv Cov (%) $\downarrow$	12.30 $\pm$ 2.71	14.02 $\pm$ 5.06	14.10 $\pm$ 4.05	7.73 $\pm$ 1.88	12.73 $\pm$ 3.17	9.40 $\pm$ 2.49	31.88 $\pm$ 4.39	16.32 $\pm$ 2.01	7.20 $\pm$ 1.70	5.46 $\pm$ 1.20	3.17 $\pm$ 0.51
Oxford Flowers (low-shot) $\rightarrow$ Deep Weeds											
ID Acc (%) $\uparrow$	98.59 $\pm$ 0.38	99.00 $\pm$ 0.57	98.89 $\pm$ 0.49	99.09 $\pm$ 0.50	98.97 $\pm$ 0.46	99.78 $\pm$ 0.15	92.38 $\pm$ 3.17	99.72 $\pm$ 0.25	100.00 $\pm$ 0.00	99.20 $\pm$ 0.51	99.59 $\pm$ 0.58
ID Cov (%) $\uparrow$	69.86 $\pm$ 2.32	63.67 $\pm$ 4.40	60.73 $\pm$ 8.23	61.70 $\pm$ 5.89	63.17 $\pm$ 6.37	64.78 $\pm$ 2.51	71.40 $\pm$ 5.39	54.20 $\pm$ 8.25	42.33 $\pm$ 5.71	51.73 $\pm$ 4.12	47.50 $\pm$ 7.44
OOD Cov (%) $\downarrow$	5.27 $\pm$ 1.26	8.87 $\pm$ 4.80	8.17 $\pm$ 1.78	8.13 $\pm$ 1.73	6.01 $\pm$ 1.34	2.34 $\pm$ 1.64	3.76 $\pm$ 2.28	6.00 $\pm$ 2.23	1.02 $\pm$ 0.41	5.61 $\pm$ 2.80	1.78 $\pm$ 1.92
Adv Cov (%) $\downarrow$	8.24 $\pm$ 2.53	13.57 $\pm$ 5.92	12.74 $\pm$ 3.86	11.56 $\pm$ 3.65	8.61 $\pm$ 3.26	1.25 $\pm$ 0.78	4.69 $\pm$ 2.90	5.55 $\pm$ 1.84	1.14 $\pm$ 0.38	6.94 $\pm$ 2.81	1.93 $\pm$ 2.19

Evaluation; Adversarial Attack Analysis:

Tested across 3 attack types:

- Gradient-based:
 - L2PGD (strong)
 - FGSM (weak)
- Non-gradient:
 - Salt-and-Pepper noise

It is also worth mentioning that all of these attacks were designed prior to C-EDL



Evaluation; Threshold Analysis:

Question:

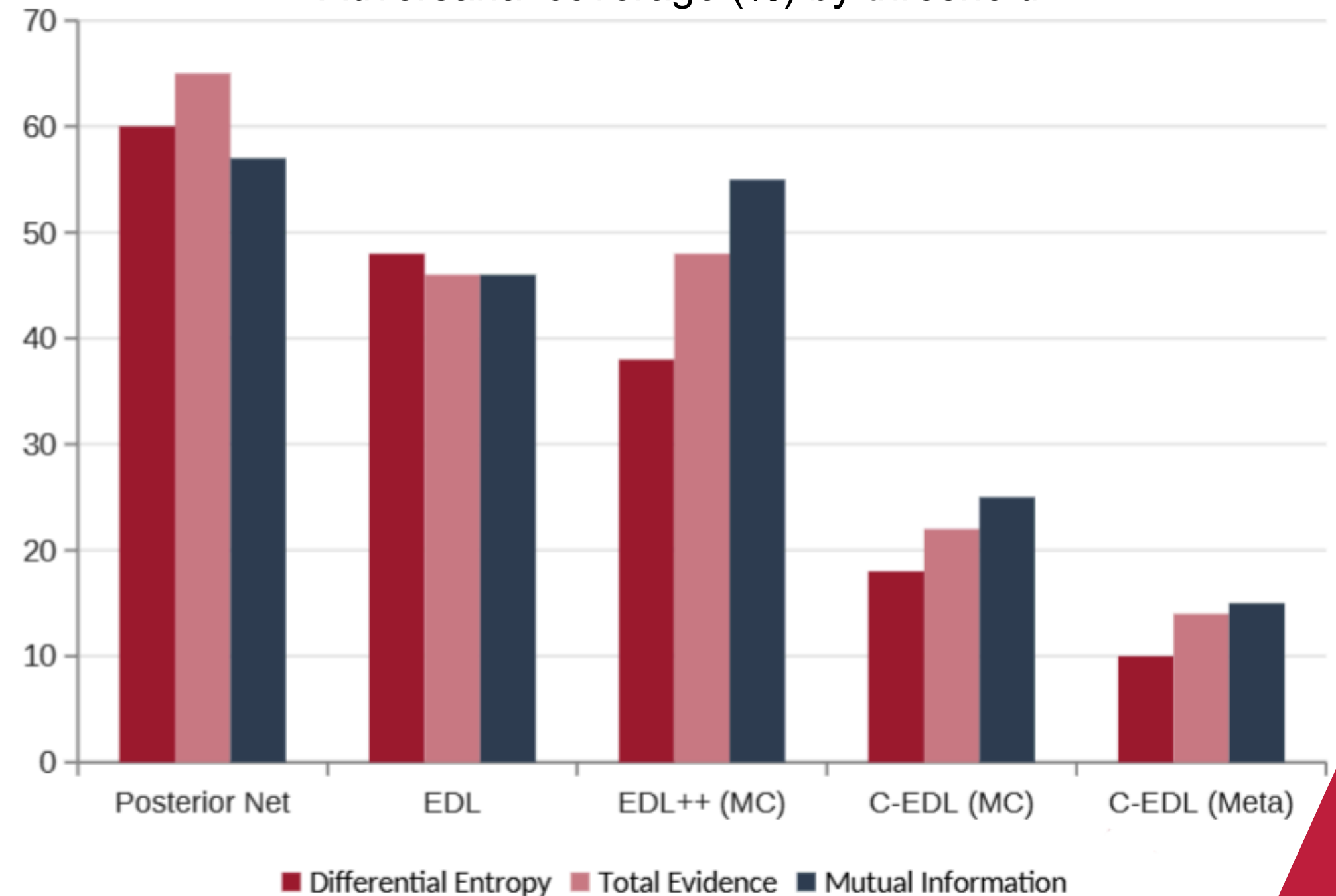
Does C-EDL's advantage depends on which ID/OOD decision you take?

- 3 threshold metrics were tested:
 - Differential Entropy
 - Total Evidence
 - Mutual Information

Findings:

- Baselines stay weak no matter the threshold
- C-EDL stayed low on all 3 metrics
- The advantage is not the threshold artifact

Adversarial coverage (%) by threshold



Personal Opinion / Critic Of the Paper:

- It only catches disagreement between views. An attack that fools the model the same way across all transformed views would slip through undetected.
- Possible circularity: the transformations tested (noise, rotation, shift) are similar to the attack types tested (noise-based attacks), so results might not generalize to very different attack styles (e.g. patch attacks).
- Since it's just a wrapper, it inherits any flaws in the underlying EDL model's evidence.

Conclusion:

Overall, C-EDL delivers a real, measurable robustness gain over prior EDL-based uncertainty methods, it:

- is a post-hoc uncertainty quantification method.
- augments any pre-trained EDL classifier with conflict analysis.
- generates label-preserving metamorphic variants of each input, quantifying intra- and inter-class disagreement.
- and detects OOD and adversarial inputs, thereby improving robustness.

Questions?



Thank you for your attention



References:

- Barker, C., Bethell, D., & Gerasimou, S. (2026). Robust Adversarial Quantification via Conflict-Aware Evidential Deep Learning. Paper presented at The Fourteenth International Conference on Learning Representations (ICLR 2026).